

NFert: An R package for nitrogen balance estimation and fertilizer management in precision agriculture

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2026-01-17

Abstract

The NFert R package provides a comprehensive toolkit for estimating nitrogen fertilization requirements in field crops. It incorporates a nitrogen balance model that accounts for factors such as soil properties, crop type, previous crops, organic amendments, and environmental conditions. The package includes functions to calculate crop nitrogen demand, soil nitrogen supply, leaching losses, immobilization, and contributions from various sources. Additionally, NFert integrates with precision agriculture techniques, enabling users to create variable rate application maps for optimizing nitrogen fertilization. This package aims to support sustainable and efficient nitrogen management practices in agriculture, aligning with the principles of integrated production and contributing to reduced environmental impact.

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1 Code metadata

The metadata for the code repository is provided in Table ??.

Table 1: Code metadata for NFert package {#tbl:metadata}

Nr	Code metadata description	Value
C1	Current code version	v1.0
C2	Permanent link to code/repository	https://github.com/mcroci/NFert
C3	Permanent link to reproducible capsule	Not available
C4	Legal code license	MIT
C5	Code versioning system used	git
C6	Software code languages, tools and services used	R
C7	Compilation requirements, operating environments and dependencies	R environment, depends on standard R packages (e.g., raster, stats)

Nr	Code metadata description	Value
C8	Link to developer documentation/manual	https://github.com/mcroci/NFert
C9	Support email for questions	michele.croci@unicatt.it

2 Motivation and significance

The efficient and sustainable management of nitrogen (N) fertilization is a critical challenge in modern agriculture. Excess nitrogen application can lead to environmental pollution, such as nitrate leaching into groundwater and emissions of greenhouse gases (e.g., nitrous oxide), while insufficient nitrogen can limit crop yields and quality, impacting food security and farmer profitability. To address these challenges, precise and science-based methods for determining optimal nitrogen fertilization rates are essential. Integrated Production systems, as defined by regional guidelines such as the “DISCIPLINARI DI PRODUZIONE INTEGRATA” in Emilia-Romagna, provide a framework for reducing the environmental impact of agriculture while maintaining productivity. Accurate nitrogen management, which considers the complex interactions between soil, plant, and environment, is a cornerstone of such systems.

The NFert R package was developed to provide a flexible and comprehensive tool for calculating nitrogen balance and fertilizer requirements in field crops, aligning with the principles of integrated production and precision agriculture. The package is based on a well-established nitrogen balance model that considers various factors affecting nitrogen availability and loss in agricultural systems, drawing upon parameters and methodologies often outlined in regional agricultural guidelines and scientific literature.

NFert addresses several key scientific and practical problems related to nitrogen management in agricultural systems:

1. **Quantification of Site-Specific Nitrogen Balance:** The package allows users to accurately estimate the nitrogen balance for a specific field and crop under local conditions. This

moves beyond generalized recommendations by incorporating site-specific data on soil properties (texture, organic matter, C/N ratio), climate (rainfall patterns influencing leaching), crop characteristics (type, expected yield, growth stage), previous cropping history, and organic fertilizer applications.

2. **Science-Based Fertilizer Recommendation:** By providing a robust nitrogen balance calculation, NFert enables the determination of the optimal supplemental nitrogen required from fertilizers. This minimizes the risk of over- or under-fertilization, leading to more efficient nutrient use, reduced waste, lower input costs, and decreased environmental pollution.
3. **Integration of Diverse Data Sources:** NFert provides a framework for integrating data from various sources, including soil analysis, historical yield data, climate records, and potentially remote sensing data, into a unified nitrogen management strategy.
4. **Support for Precision Agriculture Implementation:** The package includes functionalities specifically designed to leverage spatial data, such as NDVI maps, for estimating variable nitrogen application rates across a field. This directly supports the implementation of Variable Rate Application (VRA) strategies, enabling targeted nutrient delivery based on within-field variability.

2.1 Contribution to scientific discovery

The NFert package contributes to scientific discovery by:

- **Facilitating Mechanistic Studies:** The modular nature of the package allows researchers to isolate and study the impact of specific factors (e.g., different soil types, varying rainfall patterns, or the use of different organic amendments) on individual components of the nitrogen balance (e.g., mineralization rates, leaching losses, immobilization). This supports a deeper understanding of the processes governing nitrogen dynamics in agricultural soils.
- **Enabling Scenario Analysis:** Researchers can use NFert to simulate and compare different nitrogen management scenarios (e.g., varying fertilizer types, application timings, or rates) and evaluate their potential impact on nitrogen use efficiency, crop yield, and environmental indicators under different environmental conditions.

- **Promoting Data Integration and Model Development:** NFert provides a foundation for integrating diverse agricultural datasets and can be used as a component within more complex agro-ecosystem models aimed at simulating long-term nutrient cycling and environmental impacts.
- **Bridging Research and Practice:** By translating scientific understanding of nitrogen dynamics into a practical computational tool, NFert facilitates the adoption of more precise and sustainable nitrogen management practices by agronomists and farmers, providing a feedback loop between research and real-world application.

While specific research papers citing NFert are anticipated, its design and capabilities are directly applicable to ongoing research in nutrient management, environmental modeling of agricultural systems, and the development of precision agriculture technologies.

2.2 Experimental setting (how to use the software)

The NFert package is an R library designed for users with a working knowledge of the R programming environment. It is typically used within the standard R console, RStudio, or integrated into R scripts for automated workflows. The general experimental setting involves:

1. **Installation:** Users install the package from a CRAN-like repository or directly from the source code repository.

```
# Install from GitHub (once available on CRAN, use install.packages("NFert"))
# devtools::install_github("mcroci/NFert")
library(NFert)
```

2. **Data Acquisition:** Necessary input data are collected. This includes standard soil analysis parameters (texture, organic matter, pH, Ntot, SOM, CN, exchangeable K, available P), historical yield data, climate data (especially rainfall), details of previous crops and organic fertilizer applications, and for precision agriculture applications, spatial data like NDVI rasters. These data are typically prepared as R data structures (vectors, data frames, raster objects).

3. **Function Calls:** Users call the relevant NFert functions, passing the prepared data as arguments. For a full nitrogen balance, the `N_balance()` function is used, while individual functions like `leaching_loss()` or `soil_fertility()` can be called for specific calculations. Precision agriculture functions like `estimate_N_rate_from_calibration_curve()` take raster objects as input.
4. **Output Analysis:** The functions return R objects (e.g., lists, data frames, raster objects) containing the calculated nitrogen balance components or recommended fertilizer rates. Users then analyze, visualize, or export these outputs for decision-making or further processing (e.g., generating application maps in GIS software).

The package documentation provides detailed information on each function, including parameter descriptions, units, and examples, guiding users through the process.

2.3 Related work in literature

The nitrogen balance framework implemented in NFert is based on widely accepted principles in soil fertility and nutrient management, drawing upon models described in classical agronomy texts and contemporary research articles. The methodologies for estimating components like nitrogen mineralization, leaching, and immobilization are grounded in soil science literature and often calibrated using field experimental data. The approaches for calculating crop nitrogen demand are based on established crop physiological principles and nutrient uptake coefficients, similar to those found in regional fertilization guidelines and scientific databases.

The integration of remote sensing data for variable rate application estimation utilizes methodologies like the NDVI-based sufficiency index approach (e.g., the Holland & Schepers method) and calibration curve techniques, which are standard practices in precision agriculture research and application Holland and Schepers [2009] Schepers et al. [1996].

NFert complements other computational tools in agricultural science, including:

- **Soil process models:** More complex models simulating detailed soil biogeochemical processes. NFert provides a simplified, practical balance approach compared to these.
- **Crop growth models:** Simulation models that predict crop development and nutrient uptake over time. NFert can provide inputs or be used to validate outputs of such models.

- **GIS software and spatial analysis libraries in R:** Tools essential for processing spatial agricultural data and utilizing the variable rate application outputs from NFert.

The specific algorithms and parameters within NFert are influenced by research conducted in the Emilia-Romagna region, as reflected in the structure and content of documents like the “DISCIPLINARI DI PRODUZIONE INTEGRATA,” which provides regional coefficients and guidelines for integrated fertilization practices.

3 Software description

The NFert package is a collection of R functions designed for calculating nitrogen balance and estimating fertilizer requirements in agricultural fields. It integrates soil science, agronomy, and precision agriculture principles to provide a comprehensive toolkit for nitrogen management. The package leverages R’s capabilities for data analysis, visualization, and modeling, making it accessible to researchers, agronomists, and farmers.

The NFert package serves as a computational engine for implementing a nitrogen balance approach to fertilizer management. It provides a structured and quantitative method for assessing the nitrogen needs of a crop and determining the optimal amount of supplemental nitrogen required from fertilizers. By encapsulating complex agronomic and soil science principles into accessible R functions, NFert empowers users to move beyond generic fertilizer recommendations and adopt more precise, site-specific practices. This precision is crucial for optimizing crop production while minimizing the negative environmental consequences associated with inefficient nitrogen use. The package’s design facilitates its integration into broader agricultural data analysis workflows, including those involving spatial data and remote sensing, thereby supporting the implementation of precision agriculture strategies.

3.1 Software architecture

The NFert package follows a modular architecture, where individual functions handle specific calculations related to the nitrogen balance model. This modularity allows users flexibility in performing specific analyses or combining functions to build a complete nitrogen balance sheet. The core components of the package include:

- **Data Input and Processing Functions:** Functions designed to take raw or processed agricultural and environmental data as input. This includes functions that interpret soil properties (e.g., `tri2()` and `tri3()` for texture classification based on USDA methods, `calc_soil_group_and_id_rag()` for determining soil group and drainage index), process rainfall data, and handle crop-specific parameters.
- **Nitrogen Balance Component Functions:** A set of functions, each dedicated to calculating a specific term in the nitrogen balance equation. These include:
 - `calc_crop_N_demand()`: Calculates crop nitrogen requirement based on expected yield and crop uptake coefficients (e.g., from `uptake_table`).
 - `soil_fertility()`: Estimates soil nitrogen supply (readily available mineral N and mineralized N from organic matter) based on soil properties (Ntot, SOM, CN) and crop calendar period, using coefficients from tables like `coefN_readily` and `coefN_mineralised`.
 - `leaching_loss()`: Calculates nitrogen losses due to leaching, considering factors like winter and spring rainfall, soil drainage index (`id_rag`), and readily available nitrogen (`b1`), referencing tables like `ca.table` and `cb.table`.
 - `calc_N_immobilization_loss()`: Estimates nitrogen immobilization loss based on soil fertility (B), oxygen availability, and soil drainage index (`id_rag`), using correction factors from `ca.table` and `cb.table`.
 - `nitrogen_from_previous_crop_residues()`: Quantifies nitrogen contribution from previous crop residues based on the previous crop type, referencing values from `e.table`.
 - `organic_fertilization()`: Calculates nitrogen contribution from organic fertilizers based on the source, application frequency, and quantity, using factors from `f.table`.
 - `natural_contribution()`: Estimates natural nitrogen inputs (e.g., atmospheric deposition) based on location and crop calendar period, using data from `g.table` and `coef_time`.
- **Integrated Nitrogen Balance Function:** The `N_balance()` function serves as a wrapper, calling several of the component functions to provide a comprehensive nitrogen balance calculation based on a single set of input parameters. This function streamlines the process for users who require a full balance sheet.

- **Precision Agriculture Integration Functions:** Functions specifically designed to work with spatial data, such as raster objects containing NDVI values. `estimate_N_rate_from_calibration_curve()` implements two-point or three-point calibration to convert NDVI to N rates, while `estimate_N_rate_from_holland_schepers()` applies the Holland & Schepers algorithm. These functions enable the estimation of variable nitrogen application rates across a field.
- **Helper Functions:** Additional utility functions such as `calculate_N_fertilization()` which computes the final nitrogen fertilization requirement from a balance result, using the formula: $N_{fert} = A - B + C1 + C2 + D - E - F_{org} - G$.
- **Internal Data Tables:** The package relies heavily on internal data tables that store parameters and coefficients derived from research and regional guidelines (likely those related to the “DISCIPLINARI DI PRODUZIONE INTEGRATA”). These tables are accessed by the functions to perform the calculations and are fundamental to the package’s operation.

The architecture is designed to be flexible, allowing users to utilize individual components for specific analyses or the integrated `N_balance()` function for a complete assessment. The precision agriculture module builds upon similar underlying principles but operates on spatial data structures, providing a distinct but complementary functionality.

Figure ?? illustrates the relationships between the different functions and components of the package.

3.2 Software functionalities

The major functionalities of the NFert package provide a comprehensive toolkit for nitrogen management in integrated production and precision agriculture systems:

- **Comprehensive Nitrogen Balance Calculation:** Estimates the net nitrogen requirement of a crop by quantifying all major inputs (soil supply, previous crop residues, organic fertilizers, atmospheric deposition) and losses (leaching, immobilization) relative to the crop’s nitrogen demand.
- **Site-Specific Nutrient Management:** Enables tailored fertilizer recommendations based on detailed, site-specific data regarding soil properties, climate, and crop characteristics, moving away from generalized approaches.

- **Soil Property Characterization:** Includes tools for classifying soil texture and interpreting soil analysis results (organic matter, total nitrogen, C/N ratio, exchangeable potassium, available phosphorus) to inform nitrogen balance calculations.
- **Quantification of Nitrogen Losses:** Provides methods to estimate nitrogen losses through key pathways like leaching, influenced by rainfall and soil drainage, and immobilization by soil microorganisms.
- **Integration of Organic Inputs:** Accurately accounts for the nitrogen contribution from various organic fertilizers and residues from previous crops, promoting the efficient use of organic resources and reducing reliance on synthetic fertilizers.
- **Precision Agriculture Capabilities:** Offers functionalities to utilize remote sensing data, such as NDVI, for estimating spatially variable nitrogen application rates across a field, supporting the implementation of variable rate fertilization strategies.
- **Support for Integrated Production Compliance:** The methodologies and calculations within NFert are designed to align with the principles and requirements of integrated production guidelines, assisting users in adhering to sustainable agricultural practices.
- **Modular and Flexible Design:** The package's functions can be used independently for specific calculations or combined for a complete nitrogen balance assessment, offering flexibility for different research and application needs.

3.3 Sample code snippets analysis

Let's illustrate the usage of the package with a code snippet for calculating the nitrogen balance of a wheat crop:

```
# Load necessary libraries
library(NFert)

# Define input parameters
expected_yield_tons_ha <- 6
crop <- "Grano tenero FF (granello)"
ccp <- "Autumn-winter crop <150 days"
```

```

clay <- 10
sand <- 40
Ntot <- 1.2
SOM <- 2
CN <- 8
oxygen_availability <- "Normal"
winter_rain <- 160
start_spring_rain <- 160
prev_crop <- "Winter cereals straw removal"
source <- "Cattle slurry"
fertorg_frequency <- "every year"
forg_quantity <- 100

# Calculate nitrogen balance
n_balance_result <- N_balance(
  location = "Plain adjacent to urbanized areas",
  expected_yield_tons_ha = expected_yield_tons_ha,
  crop = crop,
  ccp = ccp,
  clay = clay,
  sand = sand,
  Ntot = Ntot,
  SOM = SOM,
  CN = CN,
  oxygen_availability = oxygen_availability,
  winter_rain = winter_rain,
  start_spring_rain = start_spring_rain,
  prev_crop = prev_crop,
  source = source,
  fertorg_frequency = fertorg_frequency,
  forg_quantity = forg_quantity
)

```

```

# Display the nitrogen balance components
print(n_balance_result)

# Calculate the final nitrogen fertilization requirement
required_N <- calculate_N_fertilization(n_balance_result)
cat("Required nitrogen fertilization:", round(required_N, 2), "kg/ha\n")

```

This snippet demonstrates how users can input relevant parameters and obtain a comprehensive nitrogen balance calculation. The `N_balance()` function serves as a convenient entry point, aggregating the results from various internal calculations to provide a complete picture of the nitrogen dynamics for the specified scenario. The output, `n_balance_result`, is a data frame containing the calculated values for each component of the nitrogen balance:

- **A:** Crop nitrogen demand (kg/ha)
- **B:** Total soil nitrogen supply (kg/ha)
- **b1:** Readily available mineral nitrogen (kg/ha)
- **b2:** Mineralizable nitrogen from soil organic matter (kg/ha)
- **C1:** Winter nitrogen leaching loss (kg/ha)
- **C2:** Spring nitrogen leaching loss (kg/ha)
- **D:** Nitrogen immobilization loss (kg/ha)
- **E:** Nitrogen from previous crop residues (kg/ha)
- **Forg:** Nitrogen from organic fertilizer (kg/ha)
- **G:** Natural nitrogen contribution (kg/ha)

The `calculate_N_fertilization()` function then computes the final nitrogen fertilization requirement using the standard formula:

$$N_{fert} = A - B + C1 + C2 + D - E - F_{org} - G$$

The clear naming of parameters and output components enhances the interpretability of the results, allowing users to understand the contribution of each factor to the overall nitrogen status and determine the required fertilizer input.

4 Illustrative examples

Beyond the comprehensive nitrogen balance calculation, NFert offers functionalities that can be used independently or as part of a precision agriculture workflow. A key aspect of precision agriculture is the ability to manage inputs variably across a field based on observed spatial variability. NFert's functions for estimating nitrogen rates from remote sensing data are particularly relevant here.

Let's consider an example where a user wants to generate a variable rate nitrogen application map for a corn field based on NDVI imagery. Assuming the user has a raster file representing the NDVI values across the field and has determined the minimum and maximum nitrogen rates based on field trials or local recommendations (e.g., 40 kg/ha for high-NDVI areas and 80 kg/ha for low-NDVI areas), they can use the `estimate_N_rate_from_calibration_curve()` function with a two-point calibration:

```
# Load the NFert package and the raster library
library(NFert)
library(raster)

# Load the NDVI raster (replace with the actual path to your NDVI file)
# For this example, let's create a dummy raster for demonstration
set.seed(123) # for reproducibility
ndvi_raster <- raster(nrows=10, ncols=10, xmn=0, xmx=10, ymn=0, ymx=10)
raster::values(ndvi_raster) <- runif(raster::ncell(ndvi_raster), min=0.3, max=0.9)

# Define minimum and maximum nitrogen rates (kg/ha)
# corresponding to max and min NDVI respectively
min_n_rate <- 40 # N rate for high NDVI (healthy, less N needed)
max_n_rate <- 80 # N rate for low NDVI (less healthy, more N needed)

# Estimate nitrogen rates using two-point calibration
n_rate_map_2pt <- estimate_N_rate_from_calibration_curve(
  raster = ndvi_raster,
  minN = min_n_rate,
```

```

maxN = max_n_rate,
calibration_type = "two-point",
plot = TRUE # Set to TRUE to visualize the calibration curve and N rate distribution
)

# The output 'n_rate_map_2pt' is a raster layer with estimated nitrogen
# application rates in kg/ha. This raster can then be exported (e.g.,
# using raster::writeRaster) and used in precision agriculture software
# to create a variable rate application map for a fertilizer spreader.
# raster::writeRaster(n_rate_map_2pt, "n_rate_map_2pt.tif",
#
format="GTiff", overwrite=TRUE)

```

For demonstration of the Holland & Schepers method:

```

# Estimate nitrogen rates using Holland & Schepers method
# Let's assume a base N rate of 50 kg/ha
base_n_rate <- 50

hs_result <- estimate_N_rate_from_holland_schepers(
  ndvi_raster = ndvi_raster,
  base_N_rate = base_n_rate,
  plot = TRUE # Set to TRUE to visualize diagnostic plots
)

# 'hs_result$dose_raster' contains the estimated N rates based on the H&S method.
# 'hs_result$sufficiency_index_raster' contains the calculated sufficiency index.
# These raster layers can also be exported for use in precision agriculture software.
# raster::writeRaster(hs_result$dose_raster, "hs_n_rates.tif",
#
format="GTiff", overwrite=TRUE)

```

This example illustrates how NFert can directly support precision agriculture workflows by translating remote sensing data into actionable nitrogen application maps. The ability to choose between different calibration methods (two-point, three-point, or Holland & Schepers) provides flexibility

based on the available data and preferred approach. The optional plotting functionality helps users visualize the relationship between NDVI and estimated nitrogen rates and understand the distribution of recommended rates across the field, facilitating the interpretation and validation of the results before generating application maps.

5 Impact

The NFert R package has the potential to significantly impact agricultural research and practice, particularly in the area of nitrogen management and precision agriculture.

5.1 New research questions

NFert facilitates the exploration of new research questions related to site-specific nitrogen dynamics and optimized fertilization strategies under varying environmental and management conditions. For example:

- How do the parameters of the nitrogen balance model (e.g., mineralization rates, leaching coefficients) need to be adjusted or refined for different pedoclimatic zones within a region, and how can NFert be used to validate these adjustments?
- Can NFert be integrated with seasonal climate forecasts to develop proactive, risk-adapted nitrogen fertilization plans that account for anticipated variations in rainfall and temperature?
- What is the economic and environmental effectiveness of using NFert-generated variable rate nitrogen application maps compared to traditional uniform rate applications across diverse farming systems and landscapes?
- How can NFert be used in conjunction with long-term field experiments to better understand the sustainability of different nitrogen management practices and their impact on soil health and productivity over decadal timescales?
- Can the framework within NFert be extended or adapted to model the dynamics of other essential plant nutrients (e.g., phosphorus, potassium) in integrated production systems?

5.2 Improvement of existing research

NFert improves the pursuit of existing research questions by providing a standardized, transparent, and flexible computational platform for nitrogen balance calculation. Researchers can easily implement and compare different nitrogen management scenarios, analyze the sensitivity of nitrogen balance components to various input parameters, and integrate nitrogen dynamics into broader agro-ecosystem models. The package’s modularity allows researchers to isolate and study specific processes (e.g., the impact of different organic amendments on nitrogen mineralization) in detail while using a consistent framework. The integration with spatial data and remote sensing tools enhances the ability to conduct research on precision nitrogen management and variable rate application, which are increasingly important areas of study for optimizing resource use and minimizing environmental footprint. NFert’s basis on principles reflected in regional guidelines also makes it a valuable tool for evaluating the effectiveness of such guidelines and informing their future refinement.

5.3 Changes in daily practice

NFert can change the daily practice of agronomists and potentially farmers by providing a more data-driven and site-specific approach to nitrogen fertilization. Instead of relying on generalized recommendations or empirical rules, users can calculate nitrogen requirements based on their specific field conditions, leading to more informed and precise decisions. For agronomists providing advisory services, NFert can be a valuable tool for developing tailored fertilization plans for their clients, potentially improving client outcomes and strengthening the scientific basis of their recommendations. The potential to generate variable rate application maps can enable farmers with the necessary equipment to implement precision fertilization, optimizing input use, potentially improving profitability, and reducing the risk of environmental contamination. While adoption requires access to necessary input data and familiarity with the R environment, the package provides the computational engine for implementing more sophisticated nitrogen management practices in real-world agricultural settings.

5.4 Widespread use and commercial settings

As an open-source R package, NFert is freely available, which facilitates its potential for widespread use within the agricultural research community globally and among early adopters of precision

agriculture technologies. Its utility is particularly high in regions where integrated production guidelines are promoted or required. The number of downloads from repositories like CRAN (once available) and citations in future research papers will be key indicators of its adoption. While there is no information provided about current commercial use or spin-off companies, the functionalities offered by NFert, particularly the ability to perform detailed nitrogen balance calculations and generate variable rate application maps, are directly relevant to commercial precision agriculture service providers, agricultural consultants, and input suppliers. Companies offering soil sampling, remote sensing data processing, and variable rate application services could potentially integrate NFert into their service platforms to enhance their analytical capabilities and provide more precise recommendations to their farmer clients.

6 Conclusions

The NFert R package represents a valuable and timely contribution to the field of nitrogen management in agriculture. By providing a comprehensive, flexible, and transparent tool for calculating nitrogen balance and estimating fertilizer requirements, the package empowers researchers, agronomists, and potentially progressive farmers to adopt more precise and sustainable nitrogen management practices. The package’s modular architecture facilitates both detailed investigation of specific nitrogen cycle processes and integrated assessment of the overall nitrogen balance. Furthermore, the inclusion of functionalities for integrating remote sensing data underscores its relevance for modern precision agriculture applications, enabling the generation of variable rate fertilization maps. By aligning with the principles of integrated production and drawing upon methodologies relevant to regional guidelines, NFert supports efforts to optimize agricultural productivity while minimizing the environmental footprint of nitrogen fertilization. As the package is adopted and utilized by the scientific and agricultural communities, it is expected to play an increasingly important role in advancing research on nutrient management and facilitating the implementation of data-driven, sustainable farming systems globally.

7 Acknowledgments

The authors gratefully acknowledge the support of the Department of Sustainable Crop Production at Università Cattolica del Sacro Cuore di Piacenza. We also thank the agricultural extension

services and farmers who provided feedback during the development and testing of the package. The development of NFert was supported by research activities focused on integrated production and precision agriculture in the Emilia-Romagna region.

8 References

References

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